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Do supply chain disruptions contribute to inflation?

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Abstract

This dissertation examines the effect of supply chain disruption on the unit cost of production of US-based automakers during the period 2015-2024. The Bill of Lading shipment dataset containing all shipments of nine major automotive manufacturers within this period is used. A novel theoretical framework is introduced to capture two key components of supply chain risk, supplier substitutability risk and shipment delay risk. A semantic similarity index is computed using BERT embeddings of product descriptions, from which a corresponding product importance index is derived as an extension of the model. Firm and quarter fixed effect regressions yield a robust, sizable negative coefficient on supplier substitutability, indicating that greater flexibility in sourcing significantly reduces unit production costs. The effect of shipment delays was of lower magnitude but significant, indicating that shipment delays increase unit costs of production by disrupting the production process. Bootstrapping and a placebo test are used to check for the robustness of the estimates. Finally, the results are critically discussed and compared with the observed increases in vehicle retail prices, suggesting that firms tend to pass the entire price increase after a supply chain shock to the consumer. No evidence of opportunistic pricing is found in the primary market, where firms directly regulate prices. However, suggestive evidence appears in the secondary automotive market.

Acknowledgments

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I am also grateful to the Import Yeti team and Mr. Dave Applegate for generously providing access to the Bill-of-Lading shipping dataset, which formed the foundation of this study.

Contents

1 Introduction

A more fractured geopolitical world in recent years, characterized by protectionist policies, geopolitical conflict and inflationary pressures, has prompted companies to rethink their supply chains. This reshaped global economy, marked by escalating trade barriers, has rendered supply chain risk a critical determinant of production costs for firms operating across borders [15] . In response, multinational companies are reevaluating their production processes, from materials sourcing to assembly location, which collectively influence their "price-determined risk," the risk associated with rising production costs.

Rising prices have often been attributed to supply chain shocks, among other factors such as monetary and fiscal policy or labor market shocks. Companies identify supply chain shocks as a key driver of inflationary pressure, which has triggered a cost-of-living crisis in the US and other developed economies in recent years [13]. It is unclear, however, whether the degree to which firms inflate prices is justified by the supply chain shocks they experience. Franzoni et al. (2024) [11] found that during supply chain shortages, larger firms gained market share and increased profitability, suggesting that such disruptions can allow dominant firms to raise prices beyond cost increases.

In contrast, many non-academic outlets argue against "Greedflation" as the primary cause behind the post COVID inflation wave, suggesting that supply chain disruption is the core factor behind this trend [9]. This dissertation investigates whether the observed increases in unit production costs for U.S. automakers between 2015 and 2024 are attributable to changes in supplier substitutability, the ease with which firms can switch among suppliers, or to logistics disruptions, as proxied by quarterly shipment-delay indices. By quantifying how supply-chain substitutability and shipment delays affect automakers' unit costs, this dissertation addresses directly the "greedflation" versus cost-push debate. While the findings lend weight to the view that inflationary pressures stem primarily from disrupted supply chains, they should be interpreted with caution given potential endogeneity and reverse-causality concerns as discussed in Section 3.5.

The US automotive sector is the primary focus of this paper, due to being heavily exposed to supply chain "price-determined risk". Its specialized nature limits the number of qualified suppliers globally, increasing vulnerability to substitutability risk, that is, the inability to replace suppliers in the event of a shock. Automotive manufacturers rely on intricate, interconnected supply networks spanning multiple countries, a characteristic that elevates network vulnerability. Furthermore, the widespread implementation of the "just in time" system, aimed at reducing inventory costs, renders automotive firms susceptible to delays [17]. These risks, substitutability and delay, are the two components of supply chain risk examined in this dissertation.

This dissertation is split into three sections. In *Section 1* I introduce the research question and provides a detailed theoretical framework on supply chain dynamics. Relevant literature is reviewed to justify the research question and explore alternative methods. *Section 2* contains the methodology and empirical research design. The dataset is explored, and the *model of price determined supply chain risk* is introduced. *Section 3* contains the results and a detailed analytical discussion of the model implications. It concludes by addressing the research question, summarizing key findings, and contextualizing them within broader trends. Research extensions are addressed to guide future research in the supply chain economics literature.

1.1 Literature Review

The limited research on supply chain risk and its effect on the cost of production presents an exciting opportunity to contribute to the literature. Yang et al. (2023) [26] classify most quantitative supply chain risk management methods into optimization, stochastic programming, and static network analyses; however, these linear frameworks treat the supply chain as a fixed graph and overlook dynamic, non-linear interdependencies. Hosseini et al. (2019) [12] provide a robust stochastic model, combining Bayesian disruption networks with mixed-integer programming to balance cost and resilience. This work has served as an inspiration for the approach to modeling supply chain risk in this dissertation.

Antràs et al. (2017) [3] develop a model showing that firms choose their global sourcing strategies endogenously, influenced by productivity and market-specific frictions such as fixed sourcing costs.

Their work demonstrates that sourcing decisions are interdependent, where a disruption in one country can affect input costs across the entire supply chain. This framework provides a robust explanation for why sourcing channels exhibit non-linear interactions, with complementarities or substitution effects emerging across markets. Furthermore, it explicitly models price determined risk of firms, a concept I build on this dissertation. Building on these insights, this research introduces a supplier substitutability index to quantify how easily a firm can switch between suppliers in a non-linear manner.

Theoretical backing for my results is provided in research by Antràs and Chor [2]. His research suggests that sunk costs by firms to their suppliers result in lower substitutability. This implies that specific suppliers that have incurred significant investment by firms might come with a risk premium in their substitutability index, which is not accounted for in my model and would need further research and data to estimate reliably. Nevertheless, the model presented in my methodology section has substitutability scores consistent with Antràs's work, where more specialized suppliers of higher volume have higher substitutability scores, as they are difficult to replace.

Further theoretical backing for the development of a substitutability index to capture supply chain risk is provided by Barrot & Sauvagnat (2016) [4], who found that natural disaster shocks to suppliers, affect their customers disproportionately if they were hard to replace. This directly reinforces the notion that substitutability is a key modulator of disruption impact. It also highlights network effects where a single vulnerable supplier can trigger broad production and cost consequences, aligning with the dissertation's network-based risk quantification.

In terms of the methodological approach used , this dissertation innovates by introducing a contextual similarity of products based on their description. Transformer based Large Language model BERT was utilized as described by Devlin et al. 2018 [7] due to its computational efficiency and its ability to create embeddings capturing the contextual similarity of products. This is based on the attention mechanism as described by Vaswani et. al. (2017) [25]

1.2 Theoretical Framework

A brief theoretical background is provided, outlining the key assumptions and definitions which aid in the understanding of supply chain dynamics. Supply chain price-determined risk is defined in this dissertation as the risk that arises from increased exposure to supply chain bottlenecks, which directly affects the cost of production of firms. This definition is consistent with the wider literature on the topic including Tang's definition of operational supply chain risk [19]. This dissertation aims at examining the effect of supply chain risk changes on the unit cost of production of US based automotive firms. This includes firms that have production capabilities in the US.

I break supply chain risk into two components, each of which takes advantage of a portion of the available dataset. Substitutability is defined as the ability of a firm to substitute its suppliers in the case of a shock. High substitutability implies that the firm has the ability to easily substitute suppliers and products in the event of a shock or bottleneck with their existing suppliers. A lower substitutability value implies that the firm does not have the capability to substitute for suppliers given a shock, thus they are more exposed to their suppliers. Substitutability holds on the following assumption:

Assumption 1: In the short run, firm i may reallocate suppliers and products only among its current supplier set K. Establishing relationships with new suppliers, negotiating contracts, performing quality audits, and integrating logistics, incurs substantial fixed costs and renders entry into entirely new sourcing partnerships infeasible in the short run. Accordingly, our substitutability index treats K as the exclusive universe of potential suppliers.

The second component of supply chain risk is average shipment delay risk, which is defined as the average quarterly delay across products for a firm. This is included due to the importance of the Just in Time (JIT) system in manufacturing and automotive production. Just in Time (JIT) is an inventory-management philosophy pioneered at Toyota whereby materials and components arrive at the production line exactly when needed, rather than being held as buffer stock [16]. By synchronizing purchase orders, supplier deliveries and production schedules, JIT minimizes carrying costs and waste, but it also leaves firms highly exposed to any downstream delay, since even a short disruption can immediately interrupt assembly.

The data we use consists of the Bills of Lading of every product imported by the chosen automotive firms, that was shipped in the U.S. over the period of 2015–2024. Each product has a description and a Harmonized System Code (HS code) which is a detailed classification system of shipping products, and the most widely adopted globally. Some key data assumptions are integral to this research:

Assumption 2: Terminology alignment. In this analysis, "cost of goods sold," "cost of production," and "cost of revenue" are used interchangeably to denote the manufacturer's per unit production expense. The proportion of COGS attributed to production excluding labor and admin costs is between 40%–60% of COGS [1].

Assumption 3: Comparable unit costs across regions. For large multinational automakers, the cost of revenue in North America is assumed to mirror that in Germany, since uniform retail pricing implies comparable production costs once transport and tariff effects are netted out. This allows to use the international Cost of Goods Sold COGS values to construct the outcome variable, where the domestic US ones are missing/not reported.

Assumption 4: Sales as a proxy for production. Quarterly units sold serve as a proxy for units produced, under the assumption that automotive manufacturing firms aim to minimize the error between demand and output, to minimize inventory carrying costs. This holds true given the high rate of depreciation of capital in the industry and the high inventory costs. This results to sales of automotive closely following production with minor errors [10]. I use the mean discrepancies between supply and demand to calculate production values using sales values where production values are missing.

Assumption 5: Multinational companies report their values in different currencies. I convert everything to dollars for this analysis based on the spot exchange rate at the time of the value I am converting. For example, COGS reported in Japanese Yen in 2018 Q2, will be converted to dollars given the spot exchange rate USD/JPY in 2018 Q2. I assume that this will not impact the outcome variable.

2 Methodology

2.1 Data Collection and Preparation

All code and cleaned data used in this study are available at github.com/AlexTaliotis/Supply-Chain-Risk-Dissertation. This repository contains all Jupyter notebooks, Stata do-files, and raw data needed to replicate the results. The data comprises of approximately 200,000 Bills of Lading data for nine automotive firms that have operations in the US, for the period 2015 Q1 – 2024 Q4. A Bill of Lading (BoL) is defined as "*a legal document issued by a carrier to a shipper that details the type, quantity, and destination of the goods being carried*". In practice, each data point contains firm specific data on the product shipped by a supplier to the importing firm. The key variables include Company/Importer name, Supplier Name, Destination Port, Departure Port, Arrival Date, Estimated Arrival Date, HS code, Product Description and Quantity. These variables are used to construct the indices that capture supply chain price determined risk and how they were used is explained in detail below. The data was acquired from Import Yeti for academic research purposes, which filed multiple Freedom of Information Acts with US Customs over many years. Such granular import data enables the detailed analysis of substitutability as the proposed model captures *product level substitutability.* The table below shows an example data set with 20,000 data points for Tesla. Note that Tesla, amongst other companies, has multiple subsidiaries that import under different names. There might be multiple observations for each day and some days have no observations.

Company Name	Supplier name	Destination Port	Departure Port	Arrival Date	Expected Arrival Date	HS code	Product description	Quantity
Tesla	Panasonic	Los Angeles, Ca	Pusan	2024/09/28	2024/09/27	370220	Photographic or cinematographic goods	40
Tesla manufacturing	Panasonic	Oakland, Ca	Rotterdam	2024/06/28	2024/03/28	380190	Engines, boilers	354
...	...	...	...	...	...	...	...	...
Tesla	Posco	Oakland, Ca	Yantian	2015/06/28	2015/06/28	284190	Lithium ion batteries	1876
Tesla	Dahyun	New York/Newark Area, Newark, Nj	Stadersand	2015/02/28	2015/02/21	210210	Copper alloy	34

Figure 1: BoL dataset for Tesla 2015-2024

The outcome variable is the unit cost of production per quarter and is collected through SEC filings, in particular 10K and 6K reports. This restricts our analysis to public companies that have been public since 2015 and thus have sufficient SEC filings. The unit cost of production was calculated by dividing the *automotive cost of production* by the *automotive production units per quarter*. Industry experts estimate that 40–60 percent of total costs stem from supply-chain expenses and imported components—exclusive of labor and energy, which have already been accounted for. This does not affect the variation and thus will not impact on our results but is done for clarity purposes.

$$unit\ cost\ of\ production_t = \left(\frac{Automtive\ cost\ of\ production_t}{units\ of\ cars\ produced_t}\right) \times \frac{1}{2} \tag{1}$$

2.2 Data Distribution

The distribution of the data can provide insights into the supply chain dynamics of the US automotive industry. Network analysis can reveal interesting patterns such as the high degree of centrality of key suppliers and supplier countries in terms of volume. This suggests that few key suppliers have increasing importance in the network and would reinforce the argument for adding a non-linear variable in the model. This is further supported by Antràs's research as discussed in the literature review section, who found similar non-linear patterns in the relationship of supplier and firm [2]. This is also consistent with previous research that I have conducted, exploring the distribution of various industries in terms of degree centrality by volume of shipments. Figure 6 in the Appendix displays the degree centrality scores of supplier denominated by country, indicating a high concentration of suppliers by volume of shipments in a few countries, notably China and Germany.

2.3 Supply Chain Price Determined Risk model

The following section introduces the supply chain – price determined risk model (SCPDR). This aims to capture two aspects of supply chain risk that affects costs of production, substitutability and shipping delay risk, as defined and explored in the previous sections.

2.4 Supplier shipment shares for firms:

Supplier substitutability is defined as the degree to which a firm i can replace/substitute their supplier k where $k \in K$ with another supplier in the set of suppliers K, in the event of a supply chain shock. For the purposes of the final regression, we aggregate the supplier substitutability for each individual firm $i \in I$ to get a score of supplier substitutability index of firm i. The score is relational in nature, meaning that the absolute value is arbitrary and can be compared with other values of firms derived from similar data, specific to these firms.

Consider a set of products G designated by a unique HS code (Harmonized System). G represents the universe of products that all firms $i \in I$ import. Product $g \in G$ and is one of the product imported by supplier k to firm i. The quantity of product of type g imported by supplier k at quarter t is shown in equation equation 2 and is denoted by $v_{ik}^{g}(t)$.

$$v_{ik}^{g}(t) = Q_{ik}^{g}(t) \tag{2}$$

$$V_{ik}^{g}(t) = \sum_{k=1}^{K} Q_{ik}^{g}(t) \tag{3}$$

To get the total quantity of products of type g that are supplied to firm i from all its suppliers at quarter t, we can aggregate across firm's i suppliers as shown by equation 3. From here we can compute the share of all products of type g that are supplied by supplier k to firm i at quarter t by dividing the two equations to get equation equation 4. This is significant as it denotes the dependence of firm i on supplier k to source product g and can reveal overdependence or under dependence of firms on specific suppliers for specific product sourcing.

$$s_{ik}^{g}(t) = \frac{v_{ik}^{g}(t)}{V_{ik}^{g}(t)} = \frac{Q_{ik}^{g}(t)}{\sum_{k=1}^{K} Q_{ik}^{g}(t)} \tag{4}$$

2.5 Product similarity index:

This research proposes a new, unexplored yet in economic literature method, for quantifying similarity between products designated by their unique HS code and textual description. This method can provide an insight into the degree to which firms can treat their suppliers interchangeably given the products they import from each supplier. As discussed in the "Data Collection and Preparation" section, each product g has its own unique designated HS code. Each HS code is divided into subcategories of different clusters of similar products, and they all have detailed descriptions.

The proposed method involves generating semantic product embeddings from the imported product descriptions of each firm. To achieve this, I utilize a pre-trained large language model to create meaningful embeddings. For this research, I have chosen BERT, as cited in [7], due to its strong capability in generating high-quality embeddings while maintaining relatively low computational requirements.

Consider a set of product descriptions $Desc(G)$, which consists of all the descriptions of all products that belong to the set of products G. $Desc(g) \in Desc(G)$ and denotes the description of product g. BERT first tokenizes $Desc(g)$ into a sequence of tokens $Desc(g) \mapsto$ [[CLS], T_1, T_2, …, T_n, [SEP]]. This is followed by mapping each token T_i into a vector $\mathbb{R}^d$ where $d = 782$ for the base BERT model used in this research. Multi head self-attention is then applied to produce a vector representation of the learned embeddings of product descriptions denoted by $u(g)$. For the purposes of this research, I omit the details regarding self-attention and transformers process, but I encourage readers to refer to [7] and Vaswani *et al.* (2017) [25] to better *understand the transformer mechanism and its application within BERT.*

Computing the product description similarity can be achieved through computing the cosine similarity of the vector embeddings $u(g_1)$, $u(g_2)$, …, $u(G)$. The cosine similarity between two product description representation vectors is denoted by $w_{gg'}$, where $g \neq g'$ and both belong to the set of products G.

$$w_{gg'} = \cos\big(u(g), u(g')\big) = \frac{u(g) * u(g')}{\|u(g)\| * \|u(g')\|} \tag{5}$$

We can extend this metric by computing the average similarity denoted by W_g of a product g across all the products that are contained in the product universe G. This creates a more meaningful metric of how similar a product is to its peers, effectively quantifying the degree to which it can be substituted with other products in the production process. It is important to note that W_g is already normalized given that the cosine similarity of BERT embeddings takes a value between $-1 \leq w_{gg'} \leq 1$ and averaging over $w_{gg'}$ yields a normalized similarity value W_g. Thus no further normalization is needed as this would also negatively affect the semantic embeddings created by the language model in use. This method comes with some major limitations which I hope to address in future research. [1]

$$W_g = \frac{1}{|G|-1} \sum_{\substack{g'=1 \\ g \neq g'}}^{G} w_{gg'} = \frac{1}{|G|-1} \sum_{\substack{g'=1 \\ g \neq g'}}^{G} \frac{u(g) * u(g')}{\|u(g)\| * \|u(g')\|} \tag{6}$$

2.6 Substitutability index:

We can combine the two previous indexes to form a holistic substitutability index denoted by $x_{ik}^{g}(t)$ shown in equation (7).

$$x_{ik}^{g}(t) = s_{ik}^{g}(t) * W_g = \frac{Q_{ik}^{g}(t)}{\sum_{k=1}^{K} Q_{ik}^{g}(t)} * \frac{u(g) * u(g')}{\|u(g)\| * \|u(g')\|} \tag{7}$$

This equation weights the share of product g shipped to firm i by supplier k at time t, by the similarity of product g with its peers. This effectively captures the degree to which a product-supplier pair can be substituted within a firm's supply chain, providing a nuanced measure of supply flexibility and resilience in response to disruptions or changes in supplier dynamics. Aggregating by product and by supplier and dividing by the number of firms $|K|$ and number of products $|G|$, yields the firm-specific metric $X_i(t)$ which captures the average ability of firm i to substitute its suppliers for a given quarter t. This is denoted in equation (8) and will form the basis of the regression specification.

$$X_i(t) = \frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} s_{ik}^{g}(t) * W_g = \frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} \frac{Q_{ik}^{g}(t)}{\sum_{k=1}^{K} Q_{ik}^{g}(t)} * \frac{u(g) * u(g')}{\|u(g)\| * \|u(g')\|} \tag{8}$$

$X_i(t)$ represents the key metric for the substitutability risk of firm i per quarter, related to its international supply chain. The terms $s_{ik}^{g}(t)$ sum up to 1 across k unless they enter non-linearly. Thus, a non-linear extension to the model is proposed below.

Introducing non-linearity:

The previously defined measure of substitutability $X_i(t)$ is linear in nature. However, in many supply-chain contexts, a near-monopoly supplier might pose more than a proportionate risk. For instance, shifting from 70% dependence on one supplier to 80% might be far riskier than going from 20% to 30%. This is further demonstrated by the exponential supplier degree centrality distribution described earlier, suggesting few suppliers carry disproportionately large risk to the rest. To capture this possibility, a non-linear exponent γ is introduced.

[1] The semantic embeddings do not reflect a specialized level of accuracy, given the data that BERT and other used variants of BERT were trained on. The data tends to be general and can often miss the delicacies related to production. Improving this method consists of finetuning BERT or another large language model to generate embeddings that more accurately reflect the automotive/manufacturing production process. Finetuning on data such as automotive production reports, engineering textbooks and potentially implementing multi modal learning such as videos of production processes can significantly enhance the accuracy of generated product embeddings. This would allow for the model to be expanded further, generating embeddings with respect to exactly how important a given product is in the production process and more accurately reflecting the degree of similarity between two products and thus the degree to which a firm can substitute one for the other.

I focus on the case where $\gamma > 1$ which follows economic intuition as the index increases with the concentration of suppliers. When $\gamma = 1$ the term $s^g_{ik}(t)$ is 1 and disappears from the expression thus this case would not be interesting. Let γ be a parameter that modifies the weight placed on large vs. small supplier shares. The new equation becomes:

$$X_i(t) = \frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} \left[s^g_{ik}(t) \right]^\gamma * W_g = \frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} \left[\frac{Q^g_{ik}(t)}{\sum_{k=1}^{K} Q^g_{ik}(t)} \right]^\gamma * \frac{u(g) * u(g')}{\|u(g)\| * \|u(g')\|} \quad (9)$$

$$X_i(t) = \frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} \left[s^g_{ik}(t) \right]^2 * W_g \quad (10)$$

Thus, γ directly shapes how heavily high-share suppliers' factor in the index. This non-linear parameter can be estimated using MLE (Maximum Likelihood Estimation) in future research. The interpretation of γ is illustrated in equation (11):

$$f(\gamma) = \begin{cases} \gamma > 1, & \text{Large suppliers are given more weight} \\ \gamma = 1, & \text{Suppliers shares weighted linearly, } s^g_{ik}(t) \text{ is 1} \\ \gamma < 1, & \text{Small suppliers are given more weight} \end{cases} \quad (11)$$

The non-linear coefficient γ can be interpreted as follows: A $\gamma > 1$ assigns higher weights to larger suppliers in terms of share of the total imports of firm i. This would follow traditional economic intuition as a firm relying on a few large suppliers would carry a greater risk [6]. The index becomes convex in the supplier share, exemplifying the risk related to large suppliers and minimizing the risk carried by small suppliers. As shown by equation (11) γ is set $\gamma = 2$ which is formally equivalent to a Herfindahl–Hirschman concentration index (HHI) over supplier shares. Squaring each share makes the index convex, so that doubling a supplier's share quadruples its contribution to $X_i(t)$ [14]. This weighting sharply penalizes over reliance on large suppliers and thus captures the nonlinear escalation of risk when a firm depends heavily on a few dominant sources. This is done for both simplification purposes, improving the confidence intervals of our estimation by having fewer parameters to estimate, and because it is supported by economic theory and intuition.

A $\gamma < 1$ assigns more weight to smaller suppliers, as the supplier share enters the index $X_i(t)$ as a concave function. Under this coefficient, small suppliers carrying heavy risk, thus reinforcing the scenario of an interconnected network of many small but important suppliers. This suggests that firms are more vulnerable to shocks as they depend on a large number of small suppliers. It is depicted for illustration purposes as it would not follow economic intuition and the dynamics of the industry.

2.7 Delay risk variable:

Consider a set of product arrival dates for firm i, $a_i \in A_{i,t}$ and expected arrival dates $E(a_i) \in E(A_{i,t})$. The firm specific average delay per quarter can be computed as follows:

$$d_i(t) = \sum_{a \in A_{i,t}} \frac{a_i - E(a_i)}{|A_{i,t}|} \quad (12)$$

Where $|A_{i,t}|$ represents the number of elements within the set of arrival dates $A_{i,t}$, and $d_i(t)$ illustrates the average delay that firm i experiences across all products $g \in G$ and suppliers $k \in K$. As mentioned previously, this index captures the supply chain risk that firms experience due to shipment delays, which increase their inventory costs. This is exemplified by the just in time system, adopted by the automotive industry to reduce inventory costs and keep inventory low. In further research we could consider the interaction between delays and specific products to capture the differential effect that products have on delays, but for the purposes of this research we will consider the absolute delay value across all products per quarter.

2.8 Parameter estimation:

Let i index firms with $i = 1, 2, 3 \ldots I$ and t index quarters with $t = 1, 2, 3 \ldots T$. The outcome variable in our proposed regression model is the unit cost of production that can be attributed to physical stock use, as explained in the Data collection and preparation section. The unit cost of automotive production for firm i for each quarter is denoted by $C_i(t)$ and captures price determined risk/exposure of firm i at time t to its international supply chain. The two key regressors are the previously defined indexes of supply chain risk: firm–supplier substitutability denoted by $X_i(t)$ and average shipment delay $d_i(t)$. The hypothesis testing on our coefficient is shown below:

Testing for substitutability:

$$H_0: \ \beta_1 = 0 \quad (\Delta \text{ in supplier-substitutability risk has no effect on unit cost})$$

$$H_1: \ \beta_1 \neq 0 \quad (\Delta \text{ in supplier-substitutability risk has an effect on unit cost})$$

Testing for shipment delay:

$$H_0: \ \beta_2 = 0 \quad (\Delta \text{ in avg. shipment delay has no effect on unit cost})$$

$$H_1: \ \beta_2 \neq 0 \quad (\Delta \text{ in avg. shipment delay risk has an effect on unit cost})$$

These two regressors capture different aspects of supply chain risk. The following specification is proposed to measure their effect:

$$C_i(t) = \beta_1 X_i(t) + \beta_2 d_i(t) + \mu_i + \delta_t + \Gamma\Phi_{it} + \epsilon_{i,t} \tag{13}$$

or

$$C_i(t) = \beta_1 \left[\frac{1}{|K| \times |G|} \sum_{k \in K} \sum_{g \in G} \left[s_{ik}^g(t) \right]^2 * W_g \right] + \beta_2 \left[\sum_{a \in A_{i,t}} \frac{a_i - E(a_i)}{|A_{i,t}|} \right] + \mu_i + \delta_t + \Gamma\Phi_{it} + \epsilon_{i,t} \tag{14}$$

Where μ_i and δ_t are fixed effects included to isolate the effect of the chosen variables of interest. The term μ_i captures firm-specific, time-invariant unobserved heterogeneity such as brand reputation or bargaining power. The term δ_t captures quarter-specific unobserved heterogeneity such as seasonal shocks affecting all firms in a quarter. $\Gamma\Phi_{it}$ is a set of controls including US CPI inflation, employment cost index (ECIWAG) and producer price index by industry (Iron and Steel Mills and Ferroalloy Manufacturing). These controls help isolate the effect of the international supply chain shocks by controlling for confounding factors within the US that might affect unit costs.

The coefficient β_1 captures the effect of substitutability on the cost per unit of production of firms. A negative and significant coefficient indicates that rising supplier substitutability decreases the unit cost of production for firms. Such a result would align with traditional economic theory where a more diversified portfolio of suppliers reduces supply chain risk exposure. Coefficient β_2 captures the relationship between the average delay of shipment and the unit cost of production, effectively capturing the extent to which the production process is disrupted by delayed shipments, which contribute to costs. All reported standard errors are clustered at the firm level to account for serial correlation and heteroskedasticity within each firm over time [5]. As a robustness check, we also compute bootstrap standard errors for the coefficients on X_{it} and d_{it}, following the nonparametric procedure of Efron (1979) [8].

3 Results

3.1 General Trends

The final sample size consisted of 236,242 Bills of Lading from 2015 Q1-2024 Q4 for nine automotive producers that produce in the US. These are Tesla, General Motors, Ford Motor Company, Nissan, Mazda, Toyota, Mercedes Benz, BMW and Honda.

The data suggests an average substitutability of 0.478. An average quarterly delay of 3.7 days is a reasonable estimate, but a large standard deviation suggests that disruptions and delays are frequent in our sample. The unit cost of production of $18,202 is rather high but reflects the low margins of the industry given the high costs of production in manufacturing. Taking the results outlined in Figure 2 as a sample benchmark we can observe how each firm individually deviates from the benchmark.

Variable	Mean	Standard Deviation	No. of observations
X_{it}	.4784753	.1619814	357
d_{it}	3.7304	28.65736	357
Unit cost of production	18202.62	7695.658	358

Figure 2: Data distribution statistics

Regarding substitutability, for most firms it tends to fluctuate substantially. However, for larger, more established automotive firms such as General Motors, Ford Motor Company and Toyota, their exposure to their suppliers tends to concentrate around a steady state mean where in contrast Tesla and Mazda, which are fairly new players in the industry tend to experience large swings of exposure. This could be an indication of more robust supply chains of larger, more established firms.

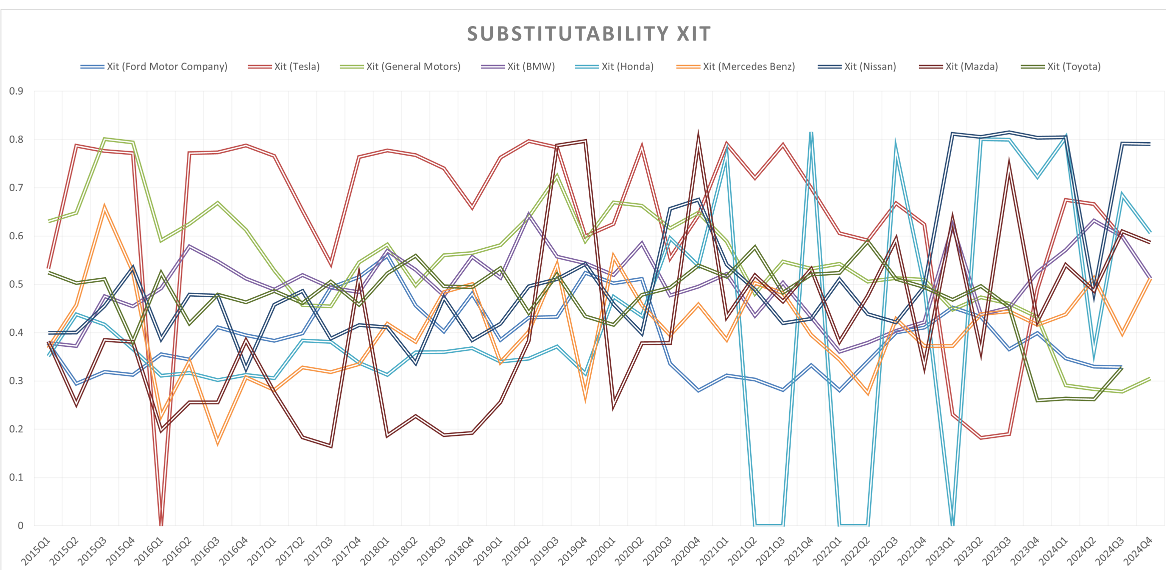

Figure 3: Substitutability across time for all firms

The average delay variable highlights a different aspect of supply chain differences across firms. Figure 4 displays the two firms with the largest average quarterly delay variation, Honda and BMW, and the firm with the lowest value Toyota. There is an observed consistency in the peaks and troughs of delays for Honda and BMW, implying common shocks that affect all shipping of certain products. The effect of the 2021 Suez Canal obstruction by Even Given is visible, as both firms experienced their largest delays that quarter, reaching a peak of 200 days delayed shipments for Honda. In contrast, Toyota is shielded from variation in average delays of shipments and is consistently close to target. This could be reflective of their much more international supply chain, since Toyota has many subsidiaries in almost all countries, which it can use to reallocate resources and avoid bottlenecks. The COGS data is displayed in Figure 7 in the Appendix.

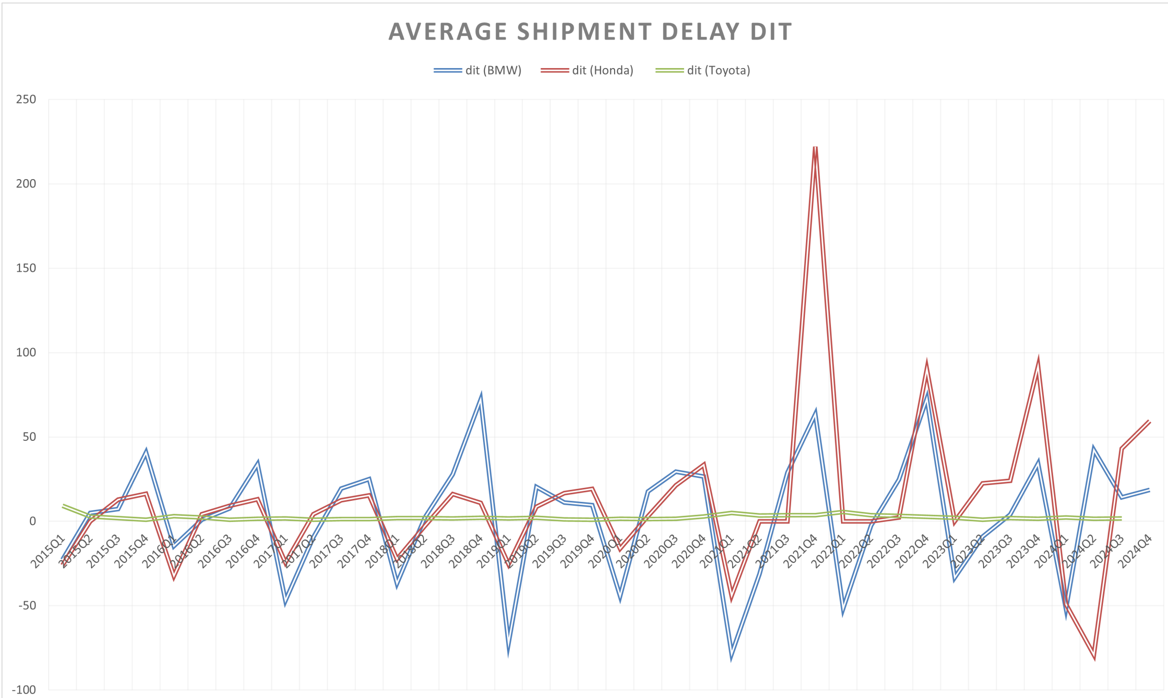

Figure 4: Average delay across time

3.2 Estimation and regression results

As described in the *Methodology* section the following specification is used to estimate the effect of substitutability and Average delays on the unit cost of production. The regression estimates are illustrated in Figure 5

$$C_i(t) = \beta_1 X_i(t) + \beta_2 d_i(t) + \mu_i + \delta_t + \Gamma\Phi_{it} + \epsilon_{i,t} \tag{15}$$

POLS no controls (1):

A pooled OLS regression without controls results to coefficient estimates of $\hat{\beta}_1 = -5715.618$ and $\hat{\beta}_2 = 24.544$, both of which are significant to the 1% level. The direction of their effect aligns with the economic intuition. A higher capability to substitute suppliers correlates with lower unit costs of production. Subsequently, increasing delays are correlated with a higher unit cost of production due to disruption in the production process. This initial step is important to confirm the direction of the effect of the two variables which align with the hypothesis discussed in the dissertation proposal.

Controls (2):

The controls included in specification (2) aim to isolate the effect of the international supply chain risk on the unit costs of production. Given that our data only includes international imports that enter the US through port, it does not account for local factors affecting costs. Domestic factors can influence costs which I aim to control for. Producer Price Index PPI (Iron and Steel Mills and Ferroalloy Manufacturing) would control for raw material price hikes that can affect costs of production within the US. The US Employment Cost Index (ECIWAG) is used to control for labor cost variation that affects costs of production locally. Finally, inflation is added as a control under the assumption that rising domestic inflation is likely to impact the costs of production. With inflation being a prominent macroeconomic figure post COVID, it could have resulted in increased costs of production for automakers in theory. Including controls does not substantially impact the size and significance of the estimates of the two key variables. This suggests that once we account for the effects of substitutability and average delays on unit costs of production, the additional controls we include do not add additional explanatory power to our regression. This reinforces the notion that the two key variables I constructed are robust predictors of unit costs of production for automakers. Furthermore, the coefficients of the controls are all insignificant, suggesting that they contribute little to quarter-to-quarter variation in automakers' unit production costs.

Quarter and firm FE (4):

In the preferred specification (4) I include both firm and quarter fixed effects, which soak up any time-invariant firm characteristics and economy-wide shocks in each quarter. Because macro controls (PPI, Employment-Cost Index, CPI) vary only at the industry or time level, they are omitted due to collinearity. In practice this means that all of their explanatory power is already absorbed by the firm and quarter fixed effect dummy variables.

The coefficient of $X_i(t)$, $\hat{\beta}_1 = -6096.117$ and is significant at the 10% level. This implies that a 1 unit increase in substitutability (from 0 to 1) results in a \$6096.117 decrease in the unit cost of production on average for automakers over the period 2015–2024, with 90% confidence that this result did not occur by chance. Equivalently, a 10% increase in substitutability of +0.1 would lead to a \$609.6 reduction in the unit cost of production. This provides evidence to accept the alternative hypothesis on the coefficient $\hat{\beta}_1$, and is proof that there is a clear link between cost of production for automakers and their ability to substitute their suppliers in the event of a shock.

The coefficient of $d_i(t)$ is estimated at $\hat{\beta}_2 = 26.861$ and is marginally insignificant at the 10% level with a p-value of $p = 0.15$. This suggests that every additional day of delay could lead to an increase in the unit cost of production of \$26.86. However, this estimate is not statistically significant at conventional levels, thus I cannot confidently distinguish it from zero. While the point estimate suggests a positive relationship, the data does not provide strong evidence that delays systematically drive up unit costs in our sample. There is an argument to be made for the significance of this estimate using standard errors obtained from bootstrapping and will be discussed in detail in the robustness test section.

The firm level dummies highlight time-invariant cost differences across firms and are illustrated in the appendix Figure 8. Taking BMW as the baseline, Toyota exhibits the lower average cost per unit at around \$14,000 below the benchmark and Ford Motor Company \$6,828 above the benchmark. This is reflected in their respective retail prices, with Toyota cars retail prices falling below the Ford Motor Company ones [18] . We cluster standard errors at the firm level to allow for within-company correlation over time; however, with only nine firms in the sample, these clustered estimates may be imprecise and hypothesis tests less reliable. Thus, robustness of the result estimates is tested using bootstrapping.

Annual and firm FE (5):

Looking at specification (5) where yearly fixed effect were implemented; we observe that the control inclusion results in delay coefficient being significant in contrast with other specifications where I cluster standard errors. This suggests that using annual dummies frees up enough residual degrees of freedom and allows the controls to enter the model, which together tightens the estimate and render the delay coefficient statistically significant.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Pooled OLS	With Controls	FE time only	FE quarter & firm	FE yearly controls	Bootstrap	No Covid – FE Quarter & firm
Substitutability (Xit)	-5715.618***	-5710.747***	-6096.117*	-6096.117*	-5554.499**	-6096.117***	-6683.275 *
	(2166.89)	(2186.84)	(2980.81)	(3019.53)	(2742.817)	(2,099.39)	(3586.899)
Delay (dit)	24.544***	22.766***	26.861	26.861	26.76576*	26.86**	27.09385
	(8.98)	(8.57)	(16.66)	(16.87)	(13.96028)	(13,508.90)	(16.6078)
PPI Control	-	14.298	-	-	11.98965	-	-
		(11.49)			(7.859367)		
ECI Control	-	302.560	-	-	-193.2678	-	-
		(206.52)			(334.9984)		
CPI Control	-	-193.578	-	-	-94.85833	-	-
		(120.27)			(171.6826)		
Time fixed effects	No	No	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	No	No	Yes	Yes	Yes	Yes
Observations	356	356	356	356	356	356	

Table 1. Effect of Supply-Chain Risk on Unit Cost
Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 5: OLS estimates of unit cost on supply-chain risk. Column (1) is baseline; (2) adds PPI, ECI, CPI controls; (3) adds time fixed effects; (4) adds firm fixed effects; (5) adds annual fixed effects and controls; (6) uses bootstrapped standard errors; (7) excludes COVID years as a placebo. Standard errors clustered at the firm level (in parentheses). *, **, *** denote$p < 0.10$, $p < 0.05$, and $p < 0.01$.

3.3 Robustness tests

Bootstrapping

To guard against small-sample bias and non-normality of our estimators in a panel of only nine firms (N=356 observations), I implement a nonparametric bootstrap with B=1 000 replications displayed in Figure 9 in the Appendix. This can be a reliable robustness test, where clustered standard errors could be biased due to small sample size. At each replication the bootstrapping method resamples the full dataset with replacement and computes the OLS estimates for $\hat{\beta}_{1b}$ and $\hat{\beta}_{2b}$. The coefficients are recorded and used to compute the bootstrapped standard errors as shown in equation 16.

$$\widehat{SE}_b(\hat{\beta}_i) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\beta}_{ib} - \bar{\beta}_i\right)^2} \tag{16}$$

The bootstrapped coefficient of $X_i(t)$ remains consistent at -6096.117, but its p-value increases to $p = 0.004$ indicating an estimated coefficient of 1% significance. This implies a robust estimate of substitutability coefficient which reinforces the negative relationship between substitutability and unit cost of production. The bootstrapped coefficient of $d_i(t)$ exhibits an increased p-value at $p = 0.047$ thus being marginally significant at the 5% level. This improved significance likely reflects the bootstrap's ability to more faithfully approximate the finite-sample distribution of our estimator, whereas the analytic cluster-robust formula may be overly conservative with so few clusters. Nonetheless, because the result lies very close to the 5% cutoff, and because several bootstrap replications failed to converge, I present both sets of confidence intervals and caution that this finding, while suggestive, is sensitive to the choice of inference method.

Placebo test - Excluding COVID

Dropping the Covid affected years and re-estimating the regression as shown by specification (7) in Figure 5 confirms the robustness of the coefficient estimates. Substitutability remains significant at the 10% level with a similar magnitude. The delay coefficient becomes marginally insignificant with a p-value of 0.103, which is closer to significance than specification (4). Thus, we can conclude that the coefficients are robust to the disproportionate effect of Covid on supply chains.

3.4 Result Analysis and Discussion

As discussed previously, the large, negative and significant coefficient of $X_i(t)$ estimated at $\hat{\beta}_1 = -6096.117$ suggests that a +0.1 increase in substitutability (10% increase) would result in a $609.61 decrease in unit costs of production for automotive firms over the period of 2015–2024. Given that the mean unit cost of production is $18,202.62, a 0.1 change in substitutability would be a sizable amount of the total unit cost of production at around 3.3% change. As observed in Figure 3 fluctuations in substitutability are often larger than 0.1 over the course of a quarter. Thus, a decrease in substitutability could yield a potential increase in the unit cost of production over a short time period. Given that the cost is not absorbed by the automaker and passed on to the consumer, this could lead to a price increase for consumers. It is important to note that a 3.3% price increase as a result of a 10% decrease in substitutability would result in a higher price increase than the CPI inflation target of 2% per year, suggesting a very real effect on consumers' welfare.

The smaller and positive coefficient of average shipment delay suggests a smaller but present effect of late shipments on the costs of production. Assuming the estimate is significant at the 5% level given the bootstrapped standard error estimates, a one day delay would result in a $26.86 increase in the unit costs of production. This can quickly accumulate during a shipping bottleneck such as the 2021 Suez Canal obstruction, where Honda experienced an almost 200 average quarter delay, which could result in a $5372 increase in unit costs of production.

This result supports the argument that there is in fact a motive to raise prices during and after periods of supply chain disruptions as a means to cover increased costs of production. A lower ability of automakers to substitute suppliers in the event of a shock will result in costs of production rising significantly. Wanting to maintain their profit margins, firms might pass part of or the whole cost to the consumer. Furthermore, shocks that impact the immediate ability of firms to produce and disrupt the Just in Time manufacturing system are likely to increase costs of production in the short term. A combination of shocks as it was the case during the COVID-19 pandemic could result in substantial surges in the costs of production of automakers. The observed 5%–10% increase in new car prices [2] during 2021–2022 is therefore not far off from what we would expect, given the combination of pandemic related shocks, a major disruption in the Suez Canal and increased geopolitical conflict that affected shipping routes all around the world [24].

3.5 Limitations and Suggestions for Further Research

The chosen methodology comes with some limitations, some of which can be addressed in future research. Reverse causality and endogeneity present a concern regarding the computation and estimation of substitutability. Often the degree to which a firm can substitute its suppliers is determined by expectations regarding future supply chain shocks. Firms tend to anticipate and to some degree react to future shocks, by either importing more in advance or reducing their imports immediately. Furthermore, endogeneity could be of concern since factors not considered in my methodology could be correlated with substitutability, thus correlating the index with the error term. This would affect our result and would not allow for causal interpretation of substitutability on costs. An instrumental variable could be used to address this source of endogeneity in future research [3].

[2] I consider new cars as auto firms are responsible for their prices in contrast with used cars whose prices are often set by independent retailers and experienced a significantly larger price hike

[3] An instrumental variable—such as quarterly country–product tariff changes on intermediate inputs, weighted by each supplier's import share—can purge endogeneity

Nonlinearities could pose a concern, both in the computation of substitutability and average delay. Both of these indices have a non-linear effect on costs, which vary over time. This is partly addressed by the $\gamma = 2$ variable when computing substitutability but estimating γ by MLE for inference—or via gradient boosting for prediction—could yield deeper insights into supplier weighting. Furthermore, delays are likely to nonlinearly affect the unit cost as they incrementally increase depreciation of machinery and could lead to temporary factory closures.

Omitted variable bias could also be of concern. Although firm and quarter fixed effects control for time-invariant heterogeneity, unobserved time-varying factors, such as shifts in firms' technology or sudden regulatory changes may still confound the relationship between our substitutability, delay indices and unit costs. Moreover, with only nine firms and 356 quarterly observations, standard "cluster-robust" errors can be downward-biased and confidence intervals overly optimistic. These concerns are partially addressed via nonparametric bootstrap inference, which recovers the finite-sample parameter distribution, and by including firm and quarter fixed effects, but the analysis could benefit from a larger sample size. Future research could be conducted by extending the sample to other manufacturing firms, enlarging the sample size and improving inference.

The main limitation of this research is the nature of the data. The outcome variable unit cost of production is computed using data from SEC filings of public firms. Given that over the last 10 years accounting standards have changed, and tend to vary by region, it can be difficult to get a consistent value for Cost of Goods Sold/Cost of Revenue/Automotive Cost of Production. Furthermore, given that different firms include varying degree of labor costs or other costs within those reported values, this can skew the results. The same is true for the units produced per quarter which are not reported consistently. Using the assumption of Demand Matching which tends to hold true we can get a good estimate of units produced per quarter from the number of cars sold per region, which closely match each other, however our estimates might lack accuracy or demand-supply mismatch scenarios that could enrich our analysis. Future research could benefit from finding an outcome variable that can be consistently observed for public and non-public companies.

Another data limitation is related to the availability and reliability of BoL shipping data. The large volume of available data makes an analysis of this form very computationally expensive and selective sampling, which is often needed, can result to selection bias. Firms import under many different names which complicates this analysis. Furthermore, firms can hide their shipping data, which they often do when involved in defense contracts or if they want to protect against competition. Lastly, human error or illegal activity could lead to measurement error or BoL that are not representative of the actual import. These limitations do not invalidate the chosen methodology but act a point of concern when interpreting coefficients causally, particularly when coefficients are marginally significant and only under certain inference conditions, as is the case with the average delay variable.

Future research suggestions

Future research should (i) extend the sample beyond automakers to other manufacturers to increase statistical power and external validity, and (ii) strengthen estimation and identification by jointly estimating nonlinear parameters via MLE or cross-validated grid search and by exploiting alternative instruments such as exchange rate shocks. As an exploratory extension, one might employ advanced forecasting methods such as VAR impulse-response or graph-neural-network models to anticipate supply-chain disruptions.

4 Conclusion

This research introduces a novel theoretical framework to compute supply chain risk and empirically tests its effect of unit costs of production. It makes use of BoL import dataset to capture the interaction between firms and their suppliers based on the products they source from each supplier. I find a negative effect of substitutability on unit costs of production for US based automakers during the period of 2015-2024, which suggests that a higher ability of a producer to substitute their suppliers in the event of a shock, corresponds to lower unit costs of production. Average shipment delay is also found to be significant [4], which corresponds to a higher unit cost of production for every additional day delayed.

The large point estimates on both substitutability and delay risk imply that the 5%–10% rise in U.S. new-car prices between 2021 and 2022 can be largely attributed to supply-chain shocks, most notably the Suez Canal blockage and COVID-19 lockdown [24]. These combined disruptions tightened critical input markets, raised production costs, and were subsequently passed on to consumers. There is indication however that the market for used cars exhibited opportunistic behavior during that period in the US, where prices surged up to 40% during late 2020-2021. This could suggest opportunistic behavior by private retailers who increase profit margins citing inflation and could support the "Greenflation" argument. This price surge is far larger than the one suggestive of our analysis and estimated coefficients and would require further investigation into the private resale market and the market for secondhand auto vehicles.

The findings of this research consist of an important contribution to the literature, as they find a strong link between supply chain risk change and cost of production. Besides addressing the "Greenflation" versus supply chain disruption argument, these findings could be incorporated by central banks or policy makers to predict sector-specific inflation following supply chain disruptions. The magnitude of the effect of supply chain risk on unit costs of production is reflective of the price surge of new cars during the periods of 2020-2021 Suez Canal blockage and COVID-19 lockdowns. This suggests that there is no evidence that US automakers act opportunistically and raise prices disproportionately during supply chain shocks, but rather that they pass the whole cost increase to the consumer.

[4]depending on the choice of inference and robustness

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A Appendix: Tables and Figures

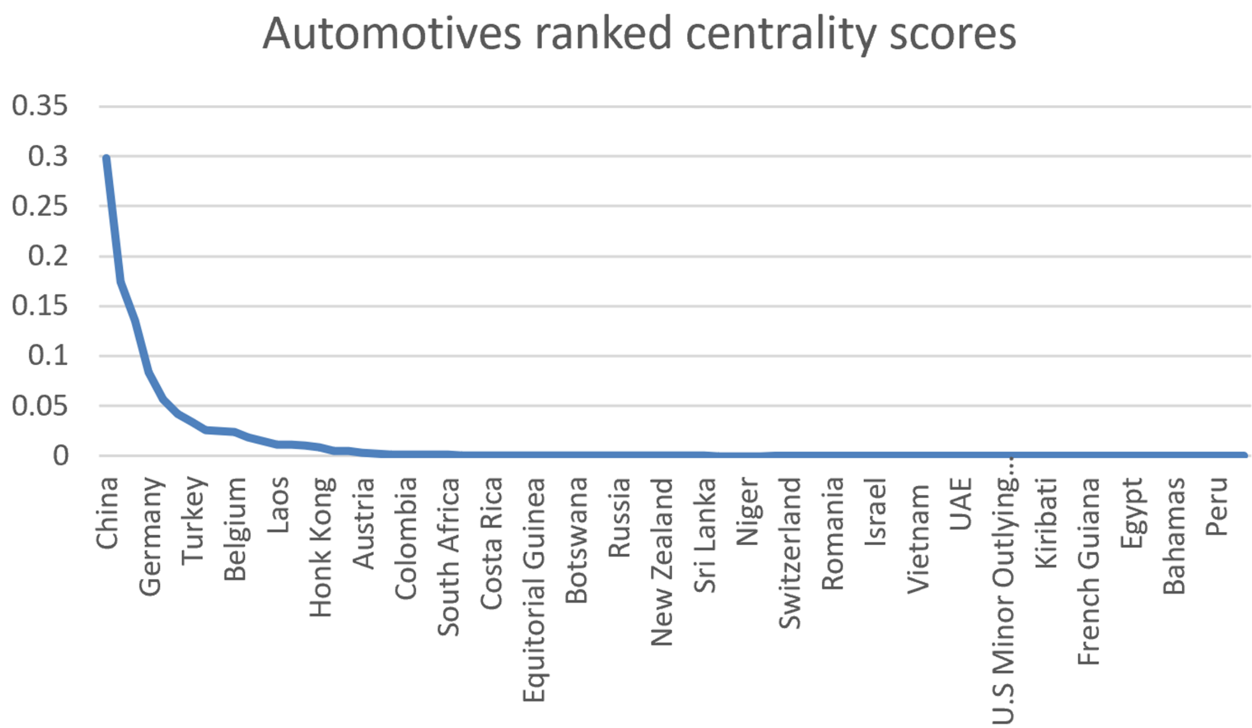

Figure 6: Automotive supplier degree centrality

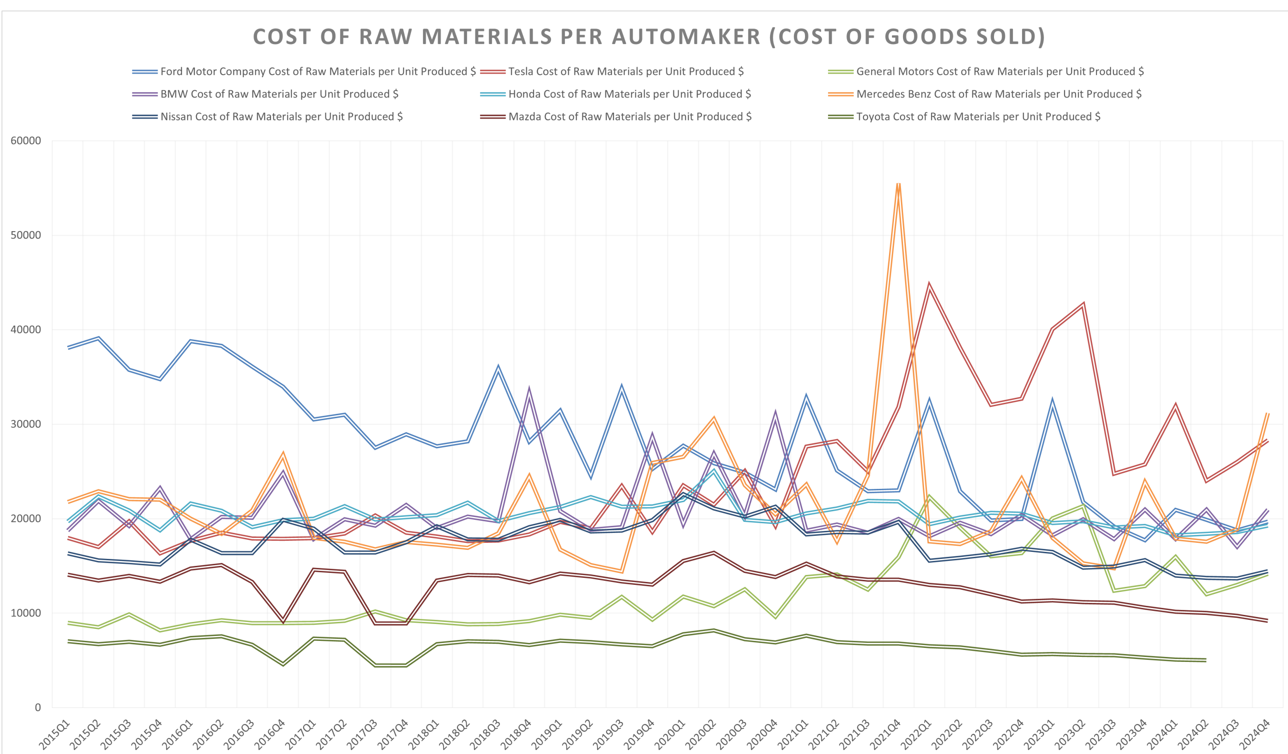

Figure 7: COGS per automaker for period 2015-2024

	(4)
	unit_cost
Xit	-6096.117*
	(3019.525)
dit	26.861
	(16.872)
BMW	0.000
	(.)
FordMotorCompany	6828.593***
	(327.234)
GeneralMotors	-8333.995***
	(153.154)
Honda	-993.182**
	(317.546)
Mazda	-8440.752***
	(261.958)
MercedesBenz	-38.767
	(293.809)
Nissan	-3127.216***
	(88.335)
Tesla	4140.470***
	(429.785)
Toyota	-14402.483***
	(70.433)
Constant	23846.442***
	(1418.068)
Observations	356

Standard errors in parentheses
Standard errors in parentheses, clustered by company_id.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 8: Regression specification (4) Quarter and Firm FE

	(1)
_bs_1	-6096.117***
	(2099.390)
_bs_2	26.861**
	(13.509)
_bs_3	1754.315***
	(182.532)
_bs_4	11.080***
	(1.601)
Observations	356

Standard errors in parentheses
Bootstrap SEs in parentheses; 1,000 reps, seed=12345.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 9: Regression specification (6) Bootstrapping estimation with 1000 replications